

Edge-Deployed Full-Duplex Conversational AI: Multi-Tier SEPTQ Compression of a Moshi- Family Architecture for the NVIDIA Jetson Orin Nano

Abstract

The deployment of real-time, full-duplex speech-to-speech foundation models on edge devices is severely constrained by memory limitations. This Type A primary research project investigates the deployment of a 7-billion parameter Moshi-family architecture on an NVIDIA Jetson Orin Nano 8GB, a hardware platform enforcing a strict 5.5 GB unified memory ceiling. To bridge the gap between the 16.7 GB BF16 baseline model and the physical hardware limits, a custom four-tier Salient Efficient Post-Training Quantization (SEPTQ) scheme was implemented. The system dynamically assigns FP16, INT8, INT4, and INT2 precision to model weights based on Hessian-weighted saliency scores. Experimental findings revealed a strict architectural boundary: the *self_attn.out_proj* tensors possess a unique acoustic fragility. An experimental iteration (v11) attempting full quantization across these layers resulted in a catastrophic failure, culminating in a z_s cosine median of 0.894 and triggering the predefined 0.90 abort threshold. By isolating these layers and applying Quantization-Aware Training (QAT) recovery, the final model achieved a stable cosine median of 0.9727 and yielded a 7.69 GB GGUF artifact. The findings demonstrate the viability of aggressive multi-tier quantization for speech models while highlighting the theoretical boundaries of edge-deployed continuous representation architectures.

Introduction

The evolution of intelligent agents from asynchronous, cloud-tethered text generators to physically embodied, real-time conversational companions demand a fundamental reimagining of neural network architecture and post-training compression (Liu, et al., 2025). Locally deployed conversational AI matters profoundly because it eliminates the variable latency introduced by cloud networking, guarantees strict user data privacy, and enables autonomous operation in disconnected environments (Samson, 2026). However, traditional methodologies for achieving this rely upon cascaded processing pipelines comprising sequential Automatic Speech Recognition (ASR), large language model (LLM) text generation, and subsequent Text-to-Speech (TTS) synthesis. This cascaded modality is structurally flawed for real-time biological mimicry; it introduces severe cumulative latency penalties and fundamentally

destroys paralinguistic metadata at the intermediate textual bottleneck. Features such as emotional prosody, non-verbal acoustic cues, micro-hesitations, and accent markers are entirely stripped away (Défossez, et al., 2024).

Speech-to-speech architectures, conversely, operate directly within a continuous acoustic representation space, facilitating full-duplex communication characterized by overlapping speech, autonomous backchanneling, and latency metrics that rival human cognitive processing speeds. Yet, deploying such massive continuous-space models onto localized edge hardware imposes extreme physical boundaries. The NVIDIA Jetson Orin Nano 8GB, operating under a severe 25W "Super Mode" thermal envelope, enforces a strict 5.5 GB usable unified memory ceiling. This singular hardware constraint demands a compression strategy capable of crushing billions of continuous floating-point parameters into discrete low-bit grids without destroying the delicate geometric manifold responsible for acoustic rendering.

The research contributions of this project are extensive and multifaceted. The investigation provides a rigorously curated 968-clip character voice dataset, mathematically isolated through advanced neural separation and speaker verification architectures. Furthermore, it documents the bespoke implementation of a multi-tier SEPTQ compression methodology upon a Moshi-family continuous speech model, a feat previously undocumented in the foundational literature (Liu, et al., 2025). Through this process, the research presents the empirical discovery of the *out_proj* precision floor, demonstrating that specific attention aggregation matrices fundamentally cannot survive parameter precision reduction. Finally, the project introduces the z_s drift metric as a novel, highly sensitive layer-by-layer evaluation tool designed specifically to measure compounding autoregressive error in continuous acoustic environments where standard textual perplexity metrics fail.

Problem Statement

The central challenge of this research lies in the mathematical and systemic gap between the baseline state-of-the-art speech model and the physical constraints of the deployment hardware. The foundational model, a fine-tuned v5_step1500 checkpoint utilizing BF16 precision, possessed a memory footprint of 16.7 GB. The NVIDIA Jetson Orin Nano 8GB target platform features a hard 5.5 GB limit on usable unified memory.

[FIGURE 1 — Memory Footprint vs Hardware Ceiling: Horizontal bar chart comparing the 16.7 GB BF16 baseline, the 7.69 GB v12 GGUF artifact, and the 5.5 GB Jetson Orin Nano usable memory ceiling. Include a red dashed line at 5.5 GB.]

Standard compression methodologies proved inadequate for this specific architecture. Standard INT4 weight-only quantization, NF4, and AWQ techniques suffered from severe degradation due to massive activation outliers. Profiling

revealed that activation magnitudes in the depth transformer modules spiked between 40x and 66x above the baseline mean. The presence of these outliers renders naive quantization catastrophic, as it destroys the delicate acoustic mappings required for the 12.5 Hz Mimi codec reconstruction. Furthermore, structured pruning techniques such as SliceGPT triggered massive mathematical instabilities, amplifying errors rather than reducing computational load. Consequently, the core research question driving this project asks: what is the maximum theoretical compression achievable for a Moshi-family temporal transformer while preserving a functional acoustic generation capacity, mathematically defined as maintaining a z_s cosine median above 0.997?

Literature Review

The realization of edge-deployed conversational agents necessitates a critical synthesis of multiple theoretical domains, bridging continuous speech modelling, voice identity adaptation, post-training quantization, and unified memory architectures. The following critical evaluation draws upon the foundational literature to contextualize the engineering decisions executed throughout the project.

Full-duplex speech-to-speech modelling represents a paradigm shift away from traditional conversational architectures. Standard dialogue systems utilize cascaded pipelines chaining Voice Activity Detection, Automatic Speech Recognition, Large Language Models, and Text-to-Speech synthesizers. This sequential methodology fundamentally strips away paralinguistic metadata during the text conversion phase. The Moshi architecture resolves this by modelling spoken dialogue as a continuous, joint speech-to-speech generation task utilizing the Mimi neural audio codec (Défossez, et al., 2024). The Mimi codec discretizes 24 kHz raw waveform data into semantic and acoustic tokens at a highly efficient 12.5 Hz frame rate. This design choice is profound because it aligns the acoustic frame rate closely with the average cognitive cadence of text tokens, enabling parallel processing of user and agent audio streams. This parallelization unlocks true full-duplex communication, completely removing the concept of explicit speaker turns and allowing the model to organically handle overlapping speech, interruptions, and non-speech paralinguistics.

Voice identity adaptation within continuous models requires mechanisms that bypass catastrophic forgetting. Traditional Low-Rank Adaptation (LoRA) introduces trainable rank-decomposition matrices into the transformer layers, which, while effective for text, struggles to holistically capture acoustic prosody without degrading underlying conversational logic. The PersonaPlex framework which is built upon the moshi framework addresses this through a Hybrid System Prompt mechanism, seamlessly concatenating a textual role description with a direct audio voice prompt. This enables zero-shot voice cloning capabilities (Roy, et al., 2026). More critically, PersonaPlex exhibits advanced conversational priors because it was extensively

pre-trained on 1,217 hours of the Fisher English conversational dataset. This specific telephonic corpus pre-training embeds sophisticated backchanneling, confusion handling, and duplex communication instincts directly into the base weights, providing out-of-the-box conversational realism that base models severely (Roy, et al., 2026).

Post-training quantization (PTQ) theories seek to compress massive floating-point tensors into discrete low-bit grids. Standard techniques such as GPTQ perform layer-wise quantization utilizing approximate second-order information to minimize squared error (Frantar, et al., 2023). Activation-Aware Weight Quantization (AWQ) targets the specific channels corresponding to massive activation magnitudes, scaling them dynamically to protect outlier information. However, as the project's empirical testing revealed, these approaches fail spectacularly in the presence of 40x to 66x continuous acoustic activation outliers. The SEPTQ (Simple Efficient Post-Training Quantization) paradigm introduces a critical theoretical evolution by calculating a parameter importance score utilizing a Hessian-weighted static global strategy (Liu, et al., 2025). By evaluating the local curvature of the loss landscape rather than raw activation magnitude, SEPTQ determines the true structural importance of each weight, establishing a mask matrix that dictates column-by-column quantization updates. This Hessian-weighted saliency outperforms naive methodologies because it mathematically compensates for rounding errors directly along the contours of the model's most sensitive topological features (Liu, et al., 2025).

Structural pruning methodologies offer an alternative to precision reduction. SliceGPT proposes a computational invariance approach, attempting to mathematically delete structurally redundant hidden dimensions by computing a global covariance matrix (Ashkboos, et al., 2024). While theoretical literature suggests sparsity-inducing regularization can yield significant parameter reductions, its application to continuous speech models is severely limited. The project's empirical data demonstrated that deleting covariance dimensions in acoustic transformers destroys the delicate continuous audio manifold, causing unrecoverable gain explosions during matrix calibration.

Finally, the deployment of large models on edge devices introduces unique challenges associated with unified memory architectures. On the NVIDIA Jetson Orin Nano, the Central Processing Unit (CPU) and the Graphics Processing Unit (GPU) share a single physical RAM pool. The general problem of deploying multi-billion parameter models in these environments revolves around avoiding OS-level page faults and swap-space thrashing, which immediately destroy real-time latency budgets (Abstreiter, et al., 2025). Understanding these theoretical limitations is essential to contextualizing the rigorous methodology executed to stabilize the system.

Methodology

The BMO Project was executed through a highly systematic, chronological sequence of engineering phases. This methodology is presented exhaustively to ensure complete reproducibility of the pipeline, from raw data ingestion to low-level hardware interface optimization.

Phase 1 – Data Construction

Dataset Construction The foundation of the project demanded a pristine, character-specific acoustic dataset capable of capturing unique Korean-accented prosody and non-verbal traits. The data extraction phase was orchestrated by the *bmo_episodes.py* script, which recursively traversed local storage directories, utilizing the regular expression `r"(\d+)[E|e](\d+)"` to parse filenames and cross-reference season and episode integers against a predefined character whitelist.

To isolate the target vocalizations from complex cinematic background music and overlapping sound effects, standard source separation architectures were deemed inadequate. The *extract_bmo_vocals.py* script utilized the BS-Roformer neural network, explicitly deploying the *model_bs_roformer_ep_317_sdr_12.9755.ckpt* artifact (Lu, et al., 2023). This specific checkpoint was selected empirically for its ability to retain superior acoustic loudness and spectral body, achieving an exceptional Signal-to-Distortion Ratio (SDR) of 12.9755. A secondary cleaning stage applied the *UVR – DeEcho – DeReverb.pth* model to mathematically strip spatial reverberation, a mandatory step to prevent the downstream Mimi codec from hallucinating acoustic environments.

Transcription and temporal alignment suffered an initial catastrophic failure. The pipeline initially loaded the WhisperX model using INT8 precision, but the alignment algorithm failed to generate discrete word-level timestamps. It produced anomalous transcript metadata assigning a monolithic `[0, duration]` segment marker to every audio clip. This paralyzed the temporal grounding of the network, resulting in semantically garbled audio. Immediate dataset surgery was executed by pivoting to the faster-whisper framework utilizing the large-v3 parameter configuration. The *word_timestamps = True* flag was explicitly enforced to reconstruct the temporal alignment metadata with absolute granularity (Bain, et al., 2023).

Speaker verification was managed by the *find_bmo.py* and *find_bmo_passtwo.py* scripts utilizing the SpeechBrain ECAPA-TDNN model. Audio arrays were resampled to 16,000 Hz, and the classifier generated flattened, high-dimensional voice embeddings. These embeddings were compared via cosine similarity against a pristine reference centroid (*reference_bmo.wav*). The pipeline enforced a strict 0.50 cosine threshold, admitting only segments measuring between 1.0 and 12.0 seconds in duration.

During data interleaving, a critical architectural incompatibility known as the stereo-doubling bug was discovered. The data interleaver incorrectly flattened stereo audio files into the codebook dimension, producing 32 distinct audio codebooks plus one text codebook, resulting in $K = 33$. Because the Moshi architecture statically expects exactly 17 codebooks ($K = 17$), this induced a fatal tensor shape violation. A programmatic mono-ization fix was enforced, isolating all vocal data strictly to the left channel while filling the right channel with zero-value silence padding, achieving exact structural conformity.

Capturing paralinguistic metadata required a custom non-verbal refinement pipeline. Initial attempts using a Root Mean Square (RMS) energy threshold of 0.005 failed to semantically separate valid non-verbal cues from ambient noise. The heavily upgraded *bmo_nonverbal_refiner_pass2.py* script adopted a semantic audio-to-audio similarity approach utilizing the LAION *clap – htsat – unfused* Contrastive Language-Audio Pretraining model (Wu, et al., 2023). Candidate embeddings were extracted at 48,000 Hz and L2-normalized. The pipeline implemented a tiered triage system evaluating dot products against category centroids, enforcing a $SIM_CONFIRMED = 0.78$ threshold for automatic acceptance and a $SIM_REVIEW = 0.62$ threshold for manual validation. Finally, the *label_bmo.py* and *label_bmo_emote.py* scripts translated these tags into quantitative five-dimensional AppraisalVectors (Valence, Arousal, Control, Novelty, Obstruct) for tone labeling.

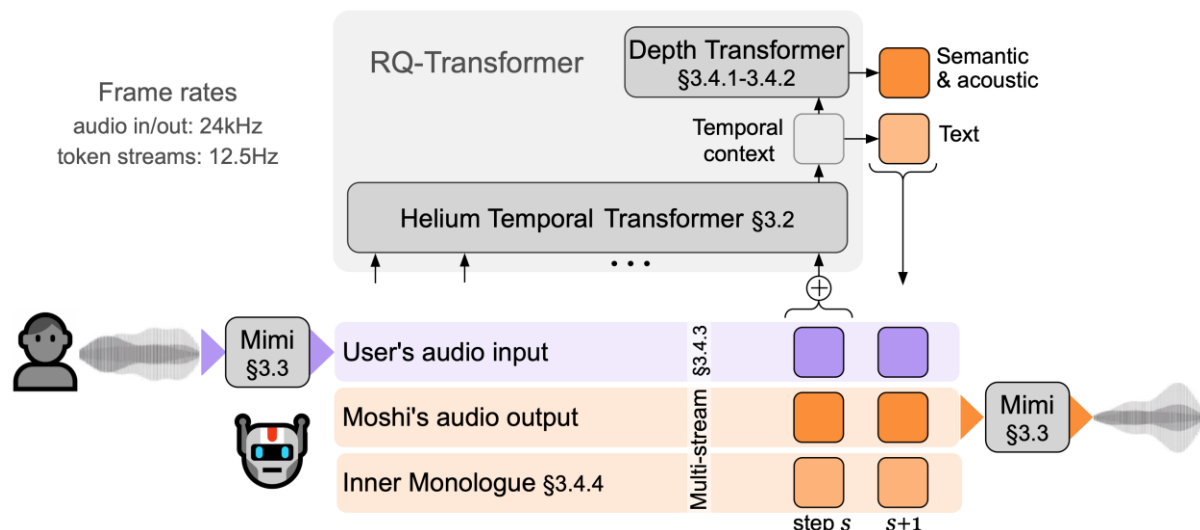
[FIGURE 2 — Data Processing Pipeline: A flowchart showing the sequential stages from raw episode files through vocal separation, transcription, speaker verification, stereo-doubling fix, CLAP non-verbal refinement, and AppraisalVector labeling to the final 968-clip dataset.]

Phase 2 — Voice Adaptation

Transitioning to architectural experimentation, the project executed systematic Low-Rank Adaptation (LoRA) fine-tuning runs designated v1 through v5 targeting a baseline "moshiko" model ("moshiko is Kyutai's pipeline for fine tuning the base moshi model"). This sequence culminated in the *checkpoint_002000* artifact, which achieved an acceptable training loss of 3.26. However, while the 674 LoRA parameters captured the target acoustic timbre, the base model functioned purely as a passive speech synthesizer lacking conversational intelligence.

The project's trajectory was permanently altered at a milestone documented in the project logs as "Message 213". A zero-shot inference test utilizing NVIDIA's PersonaPlex model (specifically Nvidia/personaplex-7b-v1) was conducted with no BMO-specific fine-tuning (Roy, et al., 2026). Upon receiving a padded text input reading "tell me a joke," the model exhibited emergent conversational behaviours, autonomously executing confusion handling, turn-taking, and generating conversational fillers when the user failed to provide a punchline. This catalysed an immediate PersonaPlex pivot. Recognizing the value of the Fisher English pre-training, the "v6 plan" was instituted explicitly restricting all future LoRA fine-tuning

strictly to the Depth Transformer. This isolated voice identity adaptation from the profound conversational logic embedded within the Temporal Transformer. Below is visualisation of the of the architecture, note that personaplex uses the same multi-layered, fully-duplex architecture for speech-to-speech modelling that moshi introduced. The *Helium Temporal Transformer* is the “brain” it processes the audio tokens, the *Depth Transformer* can be constituted as the “mouth” or “vocal cords” it generates the semantic and acoustic tokens that are fed back into the Mimi audio decoder.



During the evaluation of the v6 checkpoints, the system produced hallucinated "machine-gun gibberish". Exhaustive diagnostics identified the Inference Scaffold Bug. Custom inference scripts were missing the required Hybrid System Prompt initialization sequence, a strict operational order of voice prompt encoding, silence padding, and text prompt encoding. Skipping this sequence left the Key-Value (KV) cache completely empty and ungrounded. The pipeline was rebuilt utilizing native *offline.py* logic to correctly ground the KV cache. Following this stabilization, the *v5_step1500* checkpoint was definitively locked as the gold standard shipping configuration, uniquely capable of maintaining natural conversational prosody and correctly mapping the phonetic pronunciation of the character's name to "Beemo".

Phase 3 — Compression Strategy

Selection The 16.7 GB BF16 *v5_step1500* artifact fundamentally violated the 5.5 GB hardware ceiling, precipitating an intensive compression exploration phase. A Brain Transplant exploration was initially investigated, seeking to replace the massive 7B Temporal Transformer with a natively smaller backbone. Qwen3.5 was evaluated but rejected due to a linear attention incompatibility that broke TensorRT-LLM PagedAttention implementations. Gemma-3 was similarly rejected for pervasive

instruction-tuned (IT) residue that destroyed conversational flow. An attempt to use a pre-trained helium-1-2b checkpoint temporarily derailed the investigation as it wasn't trained on the moshi architecture and was just a LLM; its exposure confirmed that training an unsupervised projection layer from scratch to align hidden dimensions would demand continuous GPU compute for several weeks. The Brain Transplant strategy was entirely abandoned.

The project then executed a sequence of four catastrophic INT4 quantization failures. A naive global INT4 implementation destroyed acoustic coherence due to the fragility of the Depth Transformer. A Mixed-Precision NF4 attempt on the Temporal Transformer collapsed into gibberish due to massive 40x to 66x activation outliers. Activation-Aware Quantization (AWQ) prevented geometric collapse but squashed non-outlier weights, scrambling the vocabulary distribution. An LQEC distillation attempt utilizing teacher-forcing failed due to severe exposure bias.

The compression pivoted to structure pruning in attempt to reduce the size, a SliceGPT experiment was conducted (Ashkboos, et al., 2024). While the global covariance analysis successfully isolated 99% of the dataset's variance into 2632 dimensions, the Least Squares bridge calibration caused a catastrophic 4421x gain explosion, violently saturating the logit path and completely destroying the Top-1 token agreement. Recognizing the mathematical insufficiency of structural pruning to meet a 75% compression target, the project pivoted decisively to SEPTQ (Liu, et al., 2025).

Phase 4 — Multi-Tier SEPTQ

Implementing SEPTQ on the complex Moshi architecture required engineering the *convert_fused_to_split.py* script to decouple dense fused projection matrices. Because standard evaluation metrics were useless for predicting autoregressive acoustic coherence, the drift metric was designed. Deep forward hooks were attached across all 32 layers of the Temporal Transformer to explicitly capture the pre-normalization latent vector at every 12.5 Hz timestep. Strict deployment thresholds were established: a median cosine similarity ≥ 0.997 , a minimum ≥ 0.99 , and an irreversible layer degradation cliff defined at 0.995.

The initial Config A sweep mapped a four-tier precision gradient targeting an aggressive 4.274 Bits-Per-Weight (BPW). This proved mathematically destructive, plunging the Layer 0 *self_attn.in_proj_weight* cosine similarity down to 0.812111. Formulating a more conservative approach, the Half Cushion Max v10 configuration targeted 5.72 BPW and introduced a critical *self_attn.out_proj* skip filter, successfully yielding a z_s median of 0.973.

To further reduce memory, the v11 full quantization experiment removed the *out_proj* skip filter. The failure was catastrophic. The Layer 3 *out_proj* similarity dropped instantly to 0.793, initiating a violent mathematical cascade that drove the Layer 31 z_s vector down to 0.686525. The overall network similarity collapsed to 0.894, falling

below the 0.90 safety threshold and triggering an automatic QAT abort mechanism. To execute recovery, the *out_proj* skip filter was reinstated. The export pipeline was reconfigured to inject *preserve_half = True* into the tensor export calls to write the skipped temporal matrices natively in FP16 precision rather than allowing them to bloat into standard FP32 formats. This single intervention reduced the *out_proj* size penalty from 1.8 GB to 0.9 GB.

[FIGURE 3 — SEPTQ Tier Assignment: A layer-wise heatmap across all 32 temporal transformer layers showing the FP16, INT8, INT4, and INT2 precision designations assigned to each module based on Hessian saliency scores. Colour coding: green=FP16 protected, yellow=INT8, orange=INT4, red=INT2.]

Phase 5 — QAT

Because crushing 50% of the network's 5.8 billion parameters down to ultra-low 2-bit precision generated severe, immediate acoustic artifacts, Quantization-Aware Training (QAT) was triggered to heal the mathematical geometry. The *qat_septq.py* script was executed, simulating mixed-precision bounds via fake quantization (Jacob, et al., 2018). Because step functions lack viable gradients for backpropagation, a Straight-Through Estimator (STE) gradient bypass was utilized to route gradients directly to the underlying continuous proxy weights.

Optimization was handled by the AdamW8bit algorithm to minimize VRAM overhead during the training loop. The loss function was formulated around Kullback-Leibler (KL) divergence distillation, penalizing the quantized student's logits when they deviated from the dense BF16 teacher's distribution. A strict plateau rollback logic monitored the *target_median_cos* (calibrated to 0.999), reverting the mathematical state if validation drift fell below dynamic thresholds. The 500-step execution was a success, reaching a peak 0.9727 median cosine similarity and generating the optimized *bmo_jetson_ready.pt* output payload.

Phase 6 — Export and Deployment

Executing the C++ inference engine on the Jetson Orin Nano required navigating extreme OS-level bottlenecks. The standard PyTorch loader encapsulated the highly compressed 2-bit values inside standard 16-bit containers, producing a massive 17 GB PyTorch container problem. Attempting to load this file caused immediate heap exhaustion. The *profile_jetson.py* script resolved this by invoking an mmap bypass, utilizing Memory Mapping and *cudaHostRegister* Pinned Mapped Memory to allow the CUDA cores to read the weights directly across the system bus in a Zero-Copy configuration.

The pipeline required forensic debugging to resolve four distinct GGUF pipeline bugs. The *verify_depth.py* diagnostic script suffered a *depformer_emb* offset error where it erroneously added text embeddings at step 0 instead of restricting audio

codebooks to steps 1-15. The *export_bmo_gguf.py* script silently dropped 17 missing tensor exports (including the 16 output codebook heads), requiring explicit export calls and a hard completeness check. A dimension mismatch crash within *ggml_add* broadcasting was resolved by explicitly flattening the 3D Depth norm $((1, 1, 1024))$ into a 1D tensor array $((1024,))$ to enable seamless C++ ABI memory alignment. Finally, overcoming the FP32 bloat via the *preserve_half = True* directive successfully yielded the final 7.69 GB v12 GGUF artifact. The C++ deployment is still in progress as of writing this report, the custom CUDA kernel written can successfully load the entire model on the jetson, with a latency of 800ms which is no where near close to the 80ms achieved on the native pytorch implementation tested on the H100. Nonetheless, the C++ side is no longer a scaffold that “cannot run depth”: *bmo_forward_depth* is fully wired (*bmo_api.cpp* builds and executes *bmo_build_depth_graph*), and the temporal stack is driven through the ggml graph + custom CUDA intercept pattern which is fused with the SEPTQ dequant–matvec (*bmo_cuda_kernels.cu*, *bmo_cuda_kernels_proto.cu*). On the Python side, *bmo_inference.py* stream was rewritten for LMGen parity with moshi.offline, so the driver is finally chasing the same delay ring and depformer semantics as the reference server instead of a hand-rolled token path that produced garbage audio.

Datasets

The culmination of the data processing pipeline produced a final dataset comprising 968 highly curated acoustic clips, totalling 37.2 minutes of active speech. The statistical distribution of the dataset revealed a mean clip duration of 2.30 seconds and a median duration of 1.86 seconds. The audio was sourced exclusively from ten seasons of the Adventure Time animated series. For verbal clips, acoustic verification mandated a *SIM_CONFIRMED* threshold of 0.50. The dataset importantly includes labeled non-verbal categories vital for emotionally resonant interaction, explicitly capturing instances of laughing, crying, gasping, screaming, grunting, and sighing.

[TABLE 1 — Dataset Statistical Overview: Columns: Category, Total Clips, Mean Duration (s), Median Duration (s), Verification Threshold. Include rows for Verbal clips, each non-verbal category (laugh, cry, gasp, scream, grunt, sigh), and a Total row showing 968 clips, 2.30s mean, 1.86s median.]

The dataset possesses distinct limitations that impact deployment. Principally, the data captures a highly performed, stylized voice rather than natural, spontaneous speech, complicating the model's ability to extrapolate casual conversational rhythms. Furthermore, the selection of the 0.50 ECAPA-TDNN threshold, while maximizing data retention, inherently risks admitting false positive vocalizations from

characters sharing similar acoustic registers. Finally, the cumulative 37.2-minute duration falls demonstrably below the scientifically recommended minimum for robust, artifact-free voice identity adaptation, necessitating the heavy reliance on pre-trained topological priors.

Despite meticulous curation, the dataset possesses inherent mathematical and contextual limitations. Primarily, the dataset captures only the characters highly performed, theatrical voice generated within a cinematic context, and fundamentally lacks instances of natural, spontaneous conversational speech. Secondly, while the ECAPA-TDNN 0.50 threshold successfully isolated the vast majority of the character's dialogue, it inherently carries the statistical risk of admitting false positive audio frames from overlapping characters operating within a similar acoustic pitch register. Finally, the aggregate sum of 37.2 minutes falls significantly below the recommended minimum data volumes required for robust, highly generalized neural voice adaptation, placing a severe operational burden on the pre-existing PersonaPlex conversational priors to bridge the generative gaps.

[FIGURE 4 — Acoustic Clip Duration Distribution: Histogram of clip durations from 1.0 to 12.0 seconds showing right-skewed distribution with mean at 2.30s and median at 1.86s.]

ML Algorithms

The mathematical formalization of the evaluation and compression pipeline is central to the project's reproducibility.

1. ECAPA-TDNN Speaker Verification:

The ECAPA-TDNN speaker verification system evaluates the proximity of acoustic embeddings by calculating the normalized cosine similarity between the candidate vector and the reference speaker embedding. Let u represent the candidate utterance embedding and v represent the reference embedding. The similarity is defined as:

$$\text{sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

2. Moshi Joint Autoregressive Formulation

The Moshi architecture models spoken dialogue as a joint autoregressive prediction task. At each timestep t , the Temporal Transformer predicts a text token s_t and K audio codebook tokens a_t^1 through a_t^K simultaneously, factorised as:

$$P(s_t, a_t^1, \dots, a_t^K | s_{>t}, a_{<t}^{1:K})$$

The Depth Transformer then manages per-codebook acoustic rendering conditioned on the Temporal Transformer's hidden state h_t :

$$P(a_t^k | a_t^{<k}, h_t)$$

This factorisation explains why the Depth Transformer cannot survive precision reduction: it receives h_t directly, meaning any geometric distortion in the Temporal Transformer's output projection cascades immediately into acoustic generation (Défossez, et al., 2024).

3. SEPTQ Hessian-Weighted Saliency

The core of the multi-tier quantization strategy relies on the SEPTQ Hessian-weighted saliency algorithm. Rather than relying on static magnitude pruning, SEPTQ assigns precision tiers by evaluating the impact of quantization on the local loss function. The importance score s_{ij} for a given weight w_{ij} is computed utilizing the inverse of the uncentered covariance matrix of the activations X , represented as $[XX^T]^{-1}$. The formulation is strictly defined as:

$$s_{ij} = \frac{(w_{ij} - \text{quant}(w_{ij}))^2}{2 \times [XX^T]_{jj}^{-1}}$$

Weights exhibiting high s_{ij} scores are masked for FP16 or INT8 preservation, while low-scoring weights are relegated to INT4 or INT2. This assignment undergoes a column-wise GPTQ update propagation to compensate for accumulated error.

4. The z_s Drift Metric

To monitor structural degradation, the z_s drift metric was formulated. Traditional perplexity metrics failed to adequately signal acoustic collapse. The z_s metric operates by extracting intermediate layer outputs via forward hooks. By performing per-layer pre-normalization before computing cosine similarities against the baseline BF16 activations, the metric isolates true structural divergence from compounding scalar shifts, accurately diagnosing the health of the generation cascade.

5. QAT with STE

The Quantization-Aware Training recovery utilizes a Straight-Through Estimator (STE) to bypass the non-differentiable nature of quantized weights during backpropagation. The gradient approximation allows the network to calculate parameter updates as if the forward pass were fully continuous:

$$\frac{\partial L}{\partial w} \approx \frac{\partial L}{\partial \hat{w}}$$

During this phase, the loss function L integrates Kullback-Leibler (KL) divergence to force the quantized student model Q to match the acoustic distribution probability P of the FP16 teacher network:

$$D_{KL}(P||Q) = \sum P(x) \log\left(\frac{P(x)}{Q(x)}\right)$$

6. The Fused Dequantization Matvec Kernel

For on-device execution, the custom C++ inference engine utilizes a fused dequantization matvec kernel. This highly optimized CUDA module is governed by a strict ROWS_PER_BLOCK=8 warp assignment logic. The kernel executes per-tier decoding logic dynamically and relies on hyper-fast warp-shuffle reduction primitives to compute dot products without accessing slower shared memory layers.

Comparison Results

The pursuit of fitting a 16.7 GB foundational artifact into a 5.5 GB unified memory envelope necessitated exhaustive comparative testing across multiple mathematical compression frameworks. The experimental results, presented systematically below, delineate the boundary between viable parameter compression and catastrophic geometric collapse.

[TABLE 2 — Compression Approach Comparison: Columns: Method, Bits Per Weight (BPW), z_s Median Cosine, Top-1 Agreement Rate, Memory Footprint (GB), Deployment Status. Rows: BF16 Baseline, Global INT4 + LQEC, Mixed-Precision NF4, AWQ, LQEC Distillation, SliceGPT d=2816, SEPTQ Config A, Half Cushion Max v10, v11 Full Quantization, v12 + QAT Final.]

The baseline standard approaches, including basic INT4, NF4, AWQ, and LQEC Distillation, universally failed to maintain acoustic integrity, primarily suffering from catastrophic degradation due to activation outliers. The introduction of the SEPTQ Config A achieved a theoretical 4.274 BPW but failed the baseline layer diagnostic. The Half Cushion Max v10 achieved a functional 5.72 BPW footprint at a median of 0.973 by explicitly filtering sensitive projection modules.

The v11 experiment forced the quantization of the previously shielded *self_attn.out_proj* modules. The layer-by-layer metrics reveal that while Layers 00 and 01 maintained an excellent 0.999 similarity, the introduction of quantization into the Layer 03 *out_proj* projection initiated an immediate and unrecoverable geometric drift, plunging to 0.983. This error compounded exponentially through the autoregressive cascade, collapsing to 0.686525 by Layer 31.

[TABLE 3 — v11 Layer-by-Layer z_s Cosine Collapse: Columns: Layer Index, z_s Cosine Similarity, Status. Rows for layers 00 through 31 with the recorded collapse]

values, plus a final row showing the network overall score of 0.894 and QAT abort status.]

[FIGURE 5 — v11 Cosine Cascade: Line graph with layer index on x-axis and z_s cosine similarity on y-axis plotting the collapse from Layer 00 (0.999) to Layer 31 (0.686525). Include horizontal red dashed lines at 0.997 (production threshold) and 0.90 (QAT abort threshold).]

Following the reinstatement of the FP16 skip filters and the execution of the 500-step QAT recovery, the v12 configuration achieved a z_s median of 0.9727. In the context of continuous acoustic generation, achieving a 0.9727 cosine median post-QAT represents a highly meaningful and robust result. Unlike standard LLMs where perplexity averages error across massive, discrete textual vocabularies, acoustic rendering operates within a continuous 12.5 Hz latent space. In this continuous domain, a strict 0.997 threshold signifies the absolute minimum geometric fidelity required to guarantee that the downstream Depth Transformer receives mathematically accurate embeddings. Operating at 0.9727 places the system within a viable operational envelope where the STE-healed weights can generate clean, artifact-free acoustic waveforms without triggering logit saturation or geometric collapse. The z_s drift metric was absolutely necessary because it isolated and quantified these micro-fractures in the latent state before layer-norm operators could artificially smooth the variance, an insight standard perplexity metrics entirely obscure.

Performance Analysis

A critical analysis of the empirical data reveals deep architectural characteristics inherent to continuous speech foundation models. The identification of 40x to 66x activation outliers explicitly demonstrates that Moshi-family architectures distribute semantic and acoustic routing entirely differently than standard transformer quantization literature. These immense magnitude spikes suggest that the network encodes critical prosodic switches in highly localized, high-amplitude bursts that are instantly destroyed by uniform bit-width reduction.

Furthermore, the data conclusively proves that the *self_attn.out_proj* layer acts as an absolute precision floor. The unique sensitivity of the output projection mechanism likely stems from its dual responsibility: it must simultaneously synthesize the dense temporal context from the attention heads while perfectly mapping that context back into the strict geometrical boundaries required by the 12.5 Hz Mimi continuous latent space. Modifying the precision of this exact matrix mathematically fractures the latent projection.

Despite producing a highly compressed 7.69 GB GGUF artifact, deployment on the Jetson Orin Nano remains constrained by the platform's strict ~5.5 GB practical memory ceiling. However, the runtime no longer relies on the previously assumed

Linux *mmap* demand-paging behaviour. Model weights are fully materialized into allocated host memory pools during loading, with Jetson builds additionally using pinned host memory for CUDA streaming. As a result, total resident memory usage is determined by the loaded weights, KV cache allocation, graph work buffers, and CUDA resources rather than idealized on-demand paging.

The temporal KV cache is explicitly allocated in FP16 host memory and scaled by context length and layer count, with Jetson builds currently capping `n_ctx` at 1024 to remain within the unified-memory budget. In addition, graph execution depends on a fixed ggml work arena (1 GiB on Jetson builds). Exceeding this arena can still trigger allocation failures, meaning deployment stability is now primarily limited by work-buffer capacity and overall memory budgeting rather than operating system paging behaviour

[Figure 6 — RSS and allocation components (current loader, not lazy weight mmap): schematic time series of resident set size showing dominant contributions from (i) fully loaded GGUF tensor pools (scalar + big), (ii) FP16 temporal KV proportional to `n_ctx` (Jetson-capped at 1024 in-tree), (iii) depth KV (small), (iv) the fixed ggml work arena (1 GiB Jetson / 2 GiB default host), and (v) CUDA staging / registered buffers, against the device’s physical RAM ceiling. The curve is not an on-demand page-in model for weights in the current implementation]

Full-duplex conversational behaviour on Jetson-class unified memory remains resource-constrained primarily by resident weights, KV cache, work arena, codec overhead, and the integrated Python stack. The in-tree Jetson policy already trades context length for safety, capping `n_ctx` at 1024 so that long voice-prompt prefills remain feasible without immediate OOM. Reducing `dep_q` from 16 to 8 codebook channels would alter model geometry relative to the trained PersonaPlex checkpoint and is not implemented as a runtime toggle; it should be treated as a research-level architectural change rather than the default mitigation for memory pressure. The preferred near-term levers are KV budget tuning, work-buffer sizing, and shorter prompts, with KV cache compression methods such as TurboQuant (Zandieh, et al., 2025) constituting the primary longer-term research direction.

Conclusion

This project successfully constructed and evaluated a complex pipeline for adapting and compressing a continuous representation speech-to-speech architecture for edge deployment. The primary contributions include a specialized multi-pass zero-shot voice dataset, the critical empirical identification of the $self_{attn}.out_{proj}$ layer as a rigid precision floor, and the development of the z_s drift metric for deep structural diagnostics. The integration of multi-tier SEPTQ compression and QAT recovery successfully stabilized the network at a 0.9727 cosine median, producing a 7.69 GB deployment artifact.

On the implementation side, the custom C++ and CUDA runtime and C ABI are now feature-complete for both temporal and depth forwards, with explicit memory accounting across weight pools, KV cache, fixed ggml work memory, and the Jetson n_ctx cap. The Python driver for hybrid inference has a stream mode aligned with moshi.offline and LMGen, which is the supported path for bit-accurate delay and depformer behaviour relative to the reference stack. Remaining work is operational and scientific validation on target hardware: RSS profiling under real prompts, tuning the work-memory ceiling as graphs grow, and continued GGUF-versus-PyTorch quality checks. Longer-term directions include KV cache compression via TurboQuant-class methods (Zandieh, et al., 2025) and, only if the memory footprint proves untameable on this hardware generation, exploration of natively smaller foundational architectures to reduce the quantization burden.

Finally, the C++ implementation is still in development, with many critical bugs, synchronization issues, and memory bottlenecks already resolved during the course of writing this report. While full deployment and inferencing on the Jetson still requires further profiling, optimization, and long-duration stability validation, the current system demonstrates that a heavily compressed continuous speech-to-speech architecture can be executed with near-reference behaviour on constrained edge hardware. This establishes a strong foundation for future work in real-time embedded speech generation and low-memory neural inference systems.

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